Delta

$$\Delta = \frac{\Delta C}{\Delta S} = \frac{\delta C}{\delta S} \qquad \delta C = \delta S \cdot \Delta$$

$S_0 = \$100 \qquad C = \$10. \qquad \Delta = 0.6.$

$S\uparrow \rightarrow C\uparrow \rightarrow$ loss in C.
(short)

$\delta S \times \Delta$

Short 2,000 C ↓ Loss = $1 \times 0.6 \times 2000 = 1,200$

Long 1,200 S ↑ gain = $1 \times 1,200 = 1,200.$

Net = 0.

Check: S↑ by \$2.

Short. 2,000 C : Loss = $2 \times 0.6 \times 2,000 = 2,400$

Long 1,200 S : Gain = $2 \times 1,200 = 2,400$

0.

Long Call $\longrightarrow$ short S

$S\uparrow \rightarrow C\uparrow$

Short Call $\longrightarrow$ Long S.

$S\uparrow \rightarrow C\downarrow$

Long Put $\longrightarrow$ Long S

$S\uparrow \rightarrow P\downarrow$

Short Put. $\longrightarrow$ Short S.

$S\uparrow \rightarrow P\uparrow$



BSM.

	Δ.	Position in S.
Long Call.	$N(d_1) > 0$	Short $N(d_1)$ S.
Short Call	$-N(d_1) < 0$.	Long $N(d_1)$ S.
Long Put	$N(d_1) - 1 < 0$	Long $1 - N(d_1)$ S.
Short Put.	$1 - N(d_1) > 0$.	Short $1 - N(d_1)$ S.

$e^{-qT} \cdot N(d_1)$

$-e^{-qT} N(d_1)$

$e^{-qT} \cdot (N(d_1) - 1)$

$e^{-qT} \cdot (1 - N(d_1))$

$$f = (F - K)\cdot e^{-rT} = S_0 - K\cdot e^{-rT}$$

$$F = S_0 e^{rT} \qquad F\cdot e^{-rT} = S_0$$

$S \uparrow$ by \$1 $\rightarrow$ $f \uparrow$ by \$1

$\Delta = 1$ for forward.

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long forward. $+1$ short S.

short forward. -1. long S.

long.

Futures: $f = F - K = S_0 \cdot e^{rT} - K$.

$S_0 \uparrow$ by \$1 $\rightarrow$ $f \uparrow$ by e^{rT}

$\Delta = e^{rT}$ for futures.

$$F_0 = S_0 \cdot e^{(r-q)T} \qquad S_0 = F_0 \cdot e^{-(r-q)T}$$

$$f = (F-K)\cdot e^{-rT} = \left(S_0 \cdot e^{(r-q)T} - K\right)\cdot e^{-rT}$$

$$= \boxed{S_0 \cdot e^{-qT} - K \cdot e^{-rT}}$$

$S\uparrow$ by \$1 $\longrightarrow$ $f\uparrow$ by e^{-qT}

$$\Delta = e^{-qT}$$

Short. Put. Δ-neutral. Short 978k S.

~~Δ-neutral.~~ Short. 991,540. F.

Δ. for futures = $e^{(r-r_f).T} = e^{(0.06-0.075)\times\frac{11}{12}} = 0.9863$.

1 Futures. = 0.9863 S.

? 978k S.

$$? = \frac{978k}{0.9863} = 991,540.$$

Question 1

A put option on the ASX200 has an exercise price of 5,000 and maturity of 1 year. The risk-free interest rate is 7% p.a. The dividend yield on the index is 3% p.a. The volatility is 20% p.a. and the current level of the index is 5,000. A financial institution has a short position in the option.

a) Calculate the delta of a short position in this put option.

b) How can the position be made delta neutral?

c) Suppose that one hour later the index has increased to 5,150, how can delta neutrality be preserved?

$K = 5000$ $T = 1\,yr$ $r = 7\%$ $q = 3\%$

$\sigma = 20\%$ $S_0 = 5000$ Short Put.

a). $$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = 0.3.$$

$N(d_1) = 0.6179$

$$\Delta = e^{-qT} \cdot [1 - N(d_1)] = 0.371.$$

b). Short Put → Short S.

For each short put option, you should sell 0.371 index.

c). $K = 5.150.$ $T = 1\,yr$

$\Delta = 0.317.$ Short 0.317 index.

Action: Long (0.371 − 0.317) index.